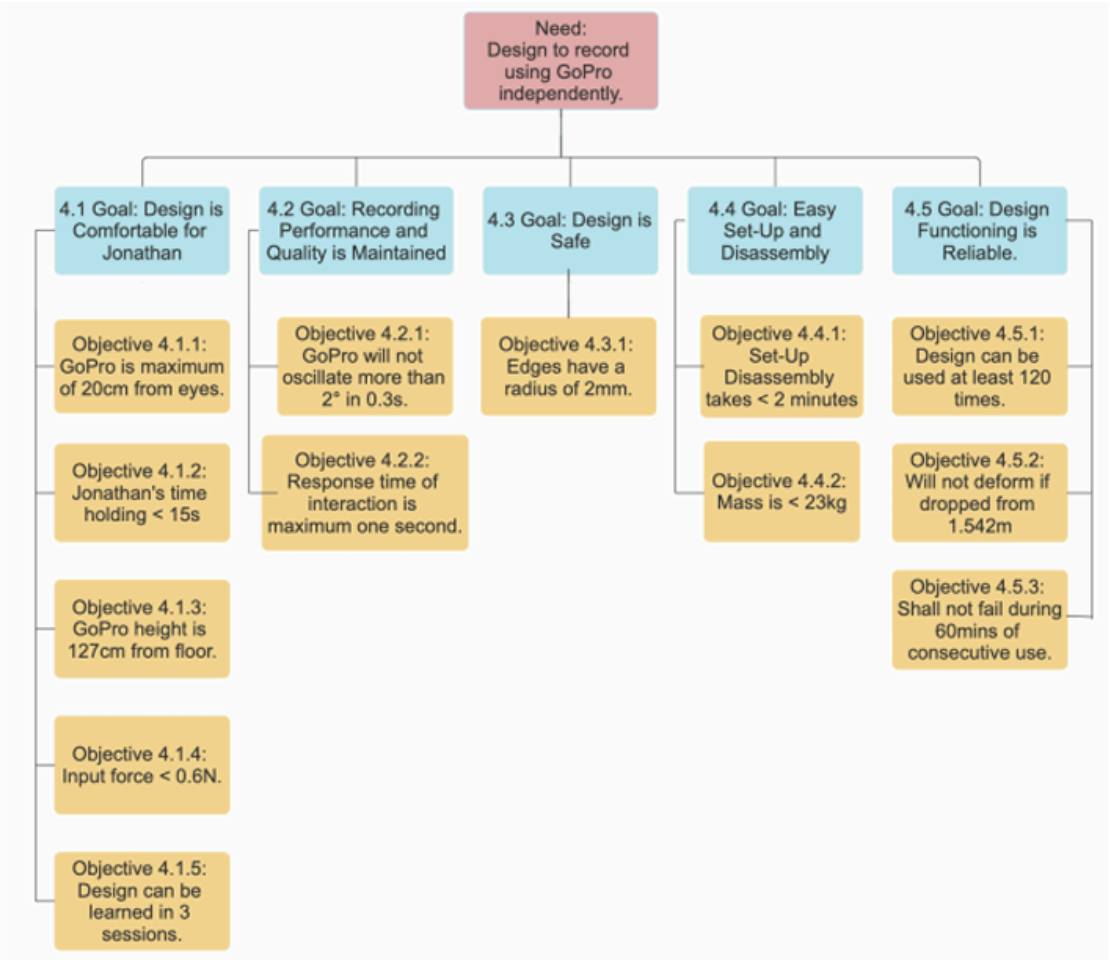
Original NGOs

From the RFP:



4.1 Jonathan's Comfort

Goal 4.1: The design does not require Jonathan to perform strenuous physical movements. Jonathan's interaction with the design should be comfortable and not exceed his physical capabilities, whilst not hindering his ability to record footage.

Objective 4.1.1: The distance between Jonathan's eyes and the GoPro will not exceed 20 cm. 20 cm between Jonathan's eyes and the GoPro buttons is optimal due to his limited vision. This was determined by finding what distance he could see keyboard keys, further evidenced in Appendix D.

Objective 4.1.2: Jonathan will never need to hold more than 200g for 15 seconds. Jonathan can comfortably hold a phone for 15 seconds in the position shown in Appendix G. The phone weighs ~200 grams, which is comparable to the GoPro's weight of 153g [4]. Therefore, to prevent strenuous movement, Jonathan will not need to hold more than 200g for 15 seconds.

Objective 4.1.3: The GoPro's default positioning will be at a vertical distance of 127 ± 11cm from the ground when Jonathan is operating the GoPro in his wheelchair.
This objective prevents Jonathan from needing to regularly unnaturally crane his head to see the touchscreen and buttons, while also fulfilling his need to record at eye level [7.13 Meeting Notes]. The calculation of this vertical distance is in Appendix D.

Objective 4.1.4: Input devices should not require more than 0.6 Newtons to successfully activate and operate.
Jonathan's physical strength limits the force with which he can operate input devices, including but not limited to buttons, dials, and switches. Testing shows he was able to easily press keyboard keys, which require 0.6 Newtons [17].

Objective 4.1.5: A maximum of 3 consecutive weekly filming sessions is needed for Jonathan to independently use the design to record, without external instruction.
The design must be intuitive and easy to learn since Jonathan is eager to film more at eye level. Since a film generally takes 2 weeks to film, he must be able to learn how to use the design within 2 weeks [7.12 Meeting Notes]. This provides a measurable benchmark for learnability of the design.

4.2 Performance and Quality of Recording

Goal 4.2: The design improves Jonathan's ability to record without hindering the filming efficiency and quality of footage taken.
This relates to ensuring Jonathan can capture footage comparable to footage recorded by Jerome within the allotted weekly filming time.
Objective 4.2.1: The GoPro must not oscillate with a frequency of more than 3 Hz at a maximum amplitude of 2° [8, 9].
Camera shake of over 2° is the minimum amplitude after which camera shake is perceived for a frequency of 3 Hz, the approximate frequency of many body movements [8, 9]. After the camera shake reaches this limit, perceived video quality decreases [10].

Objective 4.2.2: The time between triggering the recording to start and the recording starting when the design is in use shall not exceed more than 1.0 seconds [6].
Limiting response time to 1.0 seconds ensures the GoPro's responsiveness is not hindered by the design. This also supports the timely capture of Jonathan's desired scenes or moments.

4.3 Safety

Goal 4.3: The design must not be a safety hazard for Jonathan or other stakeholders at MEC.

Objective 4.3.1: The design should have a minimum radius of 2 mm for all edges and corners. 2 mm is the recommended minimum edge radius for workstation layout and creating safer human contact surfaces, but larger radii are recommended [11]. For buttons in particular however, minimum edge radius should be balanced with the ability to accurately locate and operate buttons [7.13 Meeting Notes].

4.4 Installation and Removal

Goal 4.4: The design must be easy for primary counsellors to assemble and disassemble. Assembly and disassembly includes any process before and after the design is properly set up for Jonathan to independently operate.

Objective 4.4.1: The design shall not take more than two minutes to set up or disassemble per use. Given the limited time to record in each filming session of ~45 minutes, and Jonathan's interest in recording. Since the design will allow Jonathan to accomplish similar purposes as the GoPro Extension Pole, the time of use for the design must be comparable to the Extension Pole's two minute set up and disassembly time [20].

Objective 4.4.2: Transporting the design when not in use will not weigh more than 23 kg [7]. The design must not weigh more than 23kg to enable primary counsellors to safely and efficiently transport the design.

4.5 Reliability

Goal 4.5: The design shall be reliable.
Jonathan and Jerome value reliability of the design given the frequency of filming and long duration Jonathan will likely be involved with filming at Pegasus.
Objective 4.5.1: The design shall be deployable and fully functional at least 120 times. Since filming occurs weekly and the GoPro HERO10 Black will last approximately 120 filming sessions at MEC, the design must work for at least the lifespan of the camera [Appendix D] [18, 19].

Objective 4.5.2: The design will not deform if dropped from 1.524m above the ground. The 1.524m drop height was selected as the anticipated maximum elevation of the design during set up and normal use. This prevents the design from breaking if it falls due to insecure installation or during setting up. Deformation includes fracturing, bending, and stretching [12].

Objective 4.5.3: The design shall not fail when attached to the GoPro for a minimum of 60 consecutive minutes.
Filming sessions at MEC take ~45 minutes and involve various individuals filming [7.12

Meeting Notes]. Thus, a design that does not fail for 60 minutes will be sufficient for MEC's purpose. Failing includes physical deformations and unsecure attachment of the design to the GoPro.

Reprioritization of NGOs

Priority | Major Implementation Challenges || 3 pts
Important | Thought Needed into Design || 2 pts
Preferred | Feasible for Wide Design Variety || 1 pt

(+0.5 for preference)

Goal 4.1: Comfortable Use
Objective 4.1.1: Go Pro's distance to eyes < 20 cm at all times **[KEEP]**
Reword: Visualizer screen should be between 15-25 cm from eyes at all times.
- Closer to 20 cm is better

Reword: Visualizer-eye distance varies by size of the screen due to periphery, changed to a linear function

Objective 4.1.2: Should **not** involve holding >200g for >15 seconds.
- Use of the design should involve the least possible amount of continuous holding

Objective 4.1.3: Default positioning vertical 127 ± 11cm from the ground while operating
Reword: Visualizer should be 127 ± 11cm when operating
Closer to 127 cm is better
Objective 4.1.4: Input devices need maximum 0.6 N to operate.
- Should require as less force as possible to operate

Goal 4.2: Performance and Quality of Recording
Objective 4.2.1: < 3 Hz GoPro oscillation at a maximum amplitude of 2° during use
- Should be as stable as possible

Objective 4.2.2: Actuation-response to starting/stopping record < 1.0 seconds (+0.5)

Goal 4.3: Control Unit
Objective 4.3.1: The design should have a minimum radius of 2 mm for all edges and corners.
Reword: All edges and corners should have a minimum radius of 2mm
Bigger is better
Objective 4.3.2: Full learnability within 3 weekly recording sessions (45 min each)
- As intuitive as possible

Goal 4.4: Installation/Removal
Objective 4.4.1: Maximum 2 min to set up or disassemble
- Least amount of setup-time as possible

Objective 4.4.2: Transporting the design when not in use will not weigh more than 23 kg.
- The contraption should weigh as less as possible

Goal 4.5: Reliability
Objective 4.5.1: In working condition for at least 120 cycles (filming sessions) of use.
Objective 4.5.2: The design will not deform if dropped from 1.524m above the ground.
Objective 4.5.3: GoPro is secure and undeformed for a minimum of 60 minutes.

FIRST PRIORITY: **Distance, Learnability, Assembly Time**
ENHANCE VALUES: **Maneuverability, Stabilization, Actuation-Response**
REFINEMENT: **Strength, Input Force, Durability**

Scoped/Refined NGOs

[INSERT THING]

	Obj.	Justification
Goal 1.1: Visibility	1.1.1	During the verification meeting, a member of the team guided Jonathan through multiple different monitor distances. Jonathan found a 35 cm distance to be most comfortable, far more than the 20 cm distance outlined in the RFP for a GoPro, warranting a more flexible distance calculation [See Figure A].
	1.1.2	Taken directly from the supporting RFP indicating Jonathan's eye level.
Goal 1.2: Comfort	1.2.1	Taken directly from the supporting RFP indicating strain.
	1.2.2	Changed to minimum 0.6 N as the button must not be pressed unknowingly, and added a maximum of 3 N based on _____.
Goal 1.3: Functionality	1.3.1	Taken from the widely-used 30º rule in filmmaking while allowing for tracking shots as the chair is rolled forward (see in Figure B)
	1.3.2	Taken directly from supporting RFP, rescoped to make more testable.
Goal 1.4: Implementation	1.4.1	Taken directly from supporting RFP.
	1.4.2	Taken directly from supporting RFP
	1.4.3	Taken directly from supporting RFP (?)

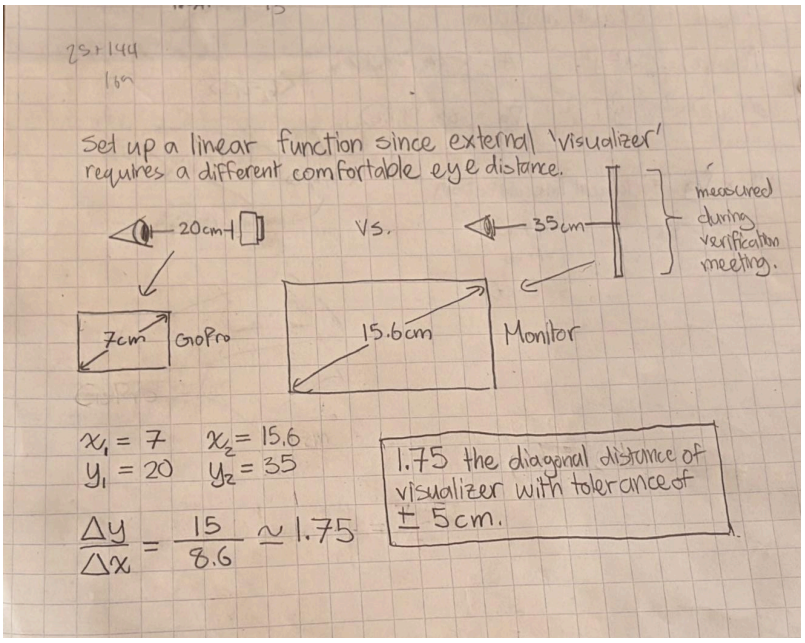


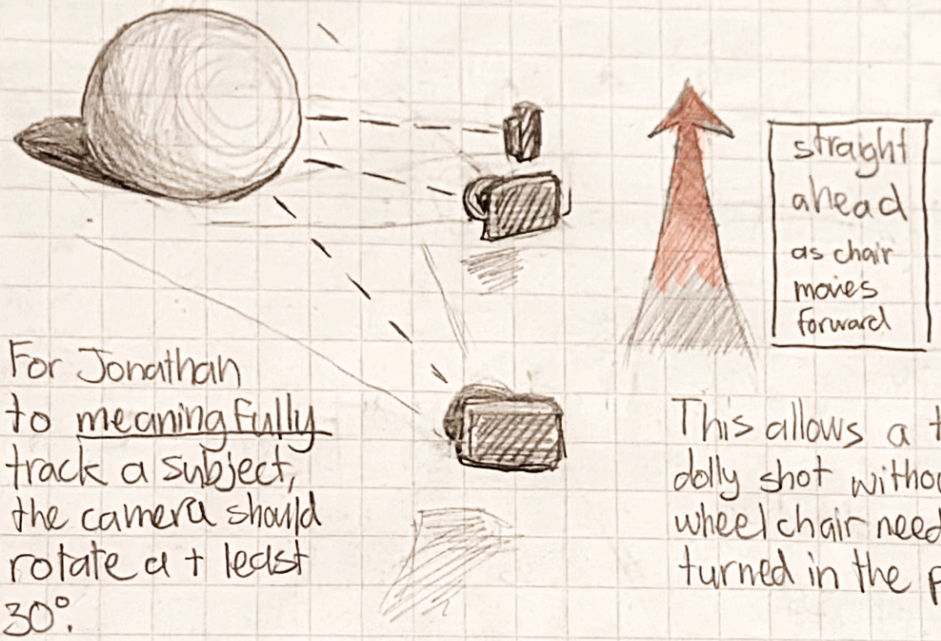
Figure A: Mathematical determination of visualizer-to-eye linear ratio, using evidence from RFP and stakeholder input

Opportunity Scoping:

MOBILITY

for shots to look different,

a minimum of 30° angle difference is required.



For Jonathan to meaningfully track a subject, the camera should rotate at least 30°.

This allows a tracking/dolly shot without the wheelchair needing to be turned in the process.

* Pan
* Tilt

For this situation, pan & tilt w/ >30° freedom unlocks the more complex but essential camera techniques in film.

Dolly — can be acheived by moving wheelchair forward/back

* Track [pictured above]

Truck — short-range workarounds, long range requires 90°

* - must be considered in design and NGOs

⊕ - available with 90° freedom (evaluation criteria)

Figure B: Filmmaking techniques in the context of Jonathan's situation

https://www.filmmakersacademy.com/glossary/30-degree-rule/

Evidence of FDCR

1.1：
To reduce visual clutter, Perkins explicitly recommends，why we hide lines

“ using plain backgrounds, minimizing clutter in the environment and materials, and increasing spacing between items.

To enhance contrast, using bright colors and high contrast to highlight targets.

To control light and glare, backlit devices can help visual attention, but also emphasise that glare can interfere with visual attention, so matte/non-glossy surfaces [note:better not to mention]should be prioritized.”

““Reducing visual clutter can improve a child’s visual attention and visual recognition.”
““be able to better see an object against a simple backdrop”

““Decreased visual clutter helps the brain organize its environment...” +”**aid visual attention and recognition**。
“

—---------- perkins

1.2: why we put the monitor in the middle, and John said his clear field is the centre.

“Place learning materials within your child's easy-to-see field of vision. Is your child's left, right, or center view clearest? If your child's lower field of vision is limited, place items on a slanted shelf with magnets or Velcro. Use dark-colored boxes to elevate items to eye level. Learning materials should be placed within the most easily visible field of vision (left, right, above, below) – using assistive devices such as slanted shelves, bookshelves, or tablet stands.”

—-----perkins

1.3 the speed of our motor is slow, less than 5w

“Consider the effect of motion. Sometimes, slow, methodical movement of objects can help maintain visual focus. However, some people with cortical visual impairment (CVI) may avoid or miss fast-moving objects.”

—----------perkins

1.4 why we use colourful buttons and shaped buttons

“Make good use of color and contrast. Use bright, eye-catching colors to attract visual attention. Take a strategic approach when using color in learning materials. Color can guide visual attention to details in objects or photographs. For example, color-coding math problems can help reduce the visual search required to find important information and solution steps. Ensure the colors enhance contrast to make the target more prominent. The key is to reduce visual searching, thereby reducing eye strain.”

—----------perkins

1.5 future steps about how to connect the table
“Use color and contrast skillfully. Use bright, eye-catching colors to attract visual attention in your environment. Color usage should be strategic—too many colors can exacerbate a feeling of clutter. For example, use bright blue tape on the rim of a container, green tape on an armrest, or yellow tape on a student's chair. Ensure the colors enhance contrast to make the target stand out more.”

—----------perkins

2.1
Why we dont choose the arm one
“When a person needs to continuously look up, extend their neck, or raise their arms, neck and shoulder discomfort and muscle strain increase; additional head/neck support may even be needed to alleviate the burden. One study directly points out that overhead work poses a significant risk to the neck and shoulder region; another study specifically developed a movable headrest because overhead work increases neck strain.”

—----------Design and ergonomic assessment of a passive head/neck supporting exoskeleton for overhead work use

3.1 future about how to choose buttons
“Regarding material selection:
The button surface material should be:
Non-slip
Not too hard or cold

And have a certain degree of tactile distinctiveness
Therefore, the most suitable surfaces are:
Matte silicone button caps
Or TPE/TPU (thermoplastic elastomer) overmolding
The surface can be finely textured to aid blind operation and prevent slippage.
“

—----------”Substitute Buttons: Exploring Tactile Perception of Physical Buttons for Use as Haptic Proxies”

4.1 why we don’t choose the arm one—for blocking the view
““Where is it best positioned for visual access of device?”
“Do they need it at a specific angle?”
“Does device **obstruct vision** when driving, communicating, watching TV or in school?”
“For outdoor use the mount should have the ability to change tilt due to glare on device from light.”

“Decrease tipping hazard by keeping device closer to the wheelbase of chair when moved out of the way.”
“Avoid **obstructing vision** when driving.”
“Static mounts may impede transfers or access.”

—----------”Mount’n Mover / *Considerations When Choosing a Mounting Device”*

5A.1 buttons if we need 5N as the requirement, radius , why not recessed, why we made arrow shaped outer face . and future steps about adding sound feedback

“Button Diameter and Operating Force: Minimum surface diameter of button controls: 13 mm. Not recessed. Operable with one hand in a closed fist position. Maximum operating force for mechanically operable parts: 19.5 N. Maximum operating force for electrically operable parts: 5 N. High-contrast, tactile, and audible confirmation information should be provided.”

—----------”A. Accessible Canada / built environment model standard”

5B.1 buttons if we need 22.2N as the requirement

“Operable parts should be usable with one hand. Tight grasping, pinching, or twisting of the wrist is not required. The force should not exceed 5 lbf = 22.2 N. Raised or flush buttons are easier to use than recessed ones. Closed-fist usability is a good usability

test.

… cannot twist a door handle, so future buttons and switches should avoid twisting/pinching actions as much as possible, prioritizing push over turn or tiny toggle.”

—----------“**Toronto Accessibility Design Guidelines”**

5C.1 buttons if we need 0.5N as the requirement

“…low force (low pressing pressure)

but not too light, otherwise it lacks tactile feedback and is prone to accidental touches.

Generally, a low force range of around 0.3–0.5 N combined with a certain travel distance is more user-friendly.

…to make the buttons:
Light force
Travel distance
Click function
Not a completely flat surface like a touchscreen.

—----------”**Push-button force-travel combinations for impaired / older users**
“（not specifically for cvi but i think they have sth in common)

6.1 why physical button

“A major advantage of physical buttons is that they are not visually dependent, are easier to identify by their shape, size, and roundness, and are safer and faster in scenarios with low visual dependence.”

—----------Substitute Buttons: Exploring Tactile Perception of Physical Buttons for Use as Haptic Proxies

7.1 why not have overhead arms
““Eye gaze systems can also impede the user’s vision for propelling the wheelchair so careful risk assessment is required when mounting eye gaze systems.” “
“This is particularly important for manual wheelchairs and with mounts that swing or

fold to the side or eye gaze systems which need to be mounted at a certain distance in front of the user.”
“consider whether the mount or device is like to get bumped or damaged, particularly if it is positioned outside the footprint of the wheelchair.”

—----------Communicate AT Mounting Assistive Technology Part 2: Considering the Abilities and Needs of the Technology User

8.1 spacing between buttons
“What to do: Ensure targets meet a minimum size or have sufficient spacing around them.”
“The target size should be at least 24 × 24 CSS pixels, or provide sufficient spacing.

A higher standard is 44 × 44 CSS pixels.”

—----------W3C

Requirements chosen for Verification (& Justification)

Requirement	Justification
Distance from eyes should be 1.75x the diagonal display size of the visualizer, with a ± 5 cm tolerance.	(Gopro size + 20cm) original RFP alongside data in Figure A
Design should have 2 axes of freedom; rotation and tilt	According to the stakeholder
Force to operate should be maximum 3 N	Force-related NGOs **must** be tested
Total Assembly Weight should be no more than 200g	Force-related NGOs **must** be tested
When in motion, design should not vibrate to the extent that recording quality is compromised	Stability affects the video quality and may compromise the safety of the design on Jonathan

Evidence of Verification/Validation

From our team, Rishana Senthan and Shupei Zhang were involved with in-person verification with Jonathan, Jerome and members of Pegasus on March 30th, 2026. The meeting can be summed up into the initial thoughts upon prototype presentation, the taking of measurements with our physical prototypes at the time, and final questions/thoughts. This heavily guided our decision making for future proxy testing and medium-fidelity/high-fidelity prototyping.

INITIAL THOUGHTS OF PEGASUS TEAM

Handlebar Design - We brought cardboard cutouts of the approximate size and shape of the handlebar to test the intuitiveness in which Jonathan can interact with the mechanisms

- Jerome commented that the design is 'very different from other team's solutions', noting it as innovative
- Elicited the greatest response from both Jerome and Jonathan
- Jerome verified that Jonathan is significantly better with thumb dexterity, the movement needed to start and stop the video via buttons as per the design
- Jerome shared worries about mounting as some parts of the wheelchair, such as the armrests, are detachable and would not prove as stable

WRC Gimbal - We brought a 3-D printed gimbal to showcase the axes of rotation and tilt, as well as a monitor and monitor arm to test screen distance.

- Jerome noted that the model could also work with the existing iPad at Pegasus
- Jonathan expressed that the concept would be "easy to use" when prompted
    - This is the only prototype of the three we presented in which Jonathan expressed probable ease of use

Table Slider - We brought a fully-functional prototype with the GoPro mount to test force and usability.

- Jonathan showed visible strain (shaking hands, numerous attempts at translation) when moving the sliders in the horizontal direction of the table, however this may be attributed to the inconsistent bearings and fastening of the slider at the time
- Jonathan required numerous prompts from Jerome to effectively translate the horizontal slider from one side of the table to the other
- Jerome commented that using the slider for long periods of time, even if the slider was a little smoother, would be tiring for Jonathan

MEASUREMENTS, TESTING & VERIFICATION

Primarily conducted for the Handlebar and WRC Gimbal designs

Handlebar Design

- Handle Grip: Jonathan was able to grab onto the cardboard handles.
- Vertical Tilt: Jonathan could extend his arms enough to achieve a tilt of over 15 degrees, the originally intended range of freedom for upwards tilt.

- Rotation: While gripping onto the handles, Jonathan was able to rotate with his arms but seemed to have difficulty turning his head to follow the movement of the GoPro slot past ± 45º (in which 0º is straight ahead)

WRC Gimbal
- Monitor Distance: 35 cm
- Monitor Arm Height: 25 cm from arm rests (tray)

FINAL THOUGHTS/QUESTIONS
- When asked directly, Jerome said the WRC Gimbal was the most feasible and low-strain, although the handlebar design was innovative
- Jonathan's favourite color was purple, and we wanted to integrate that into our designs

Validation Meeting Notes

General feedback:

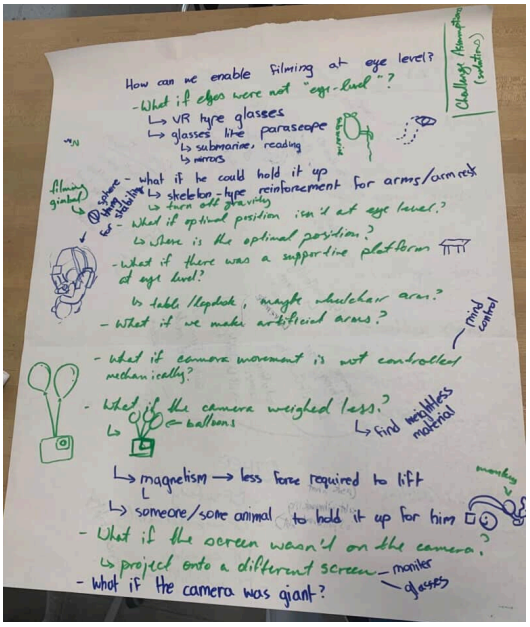
Feedback on proposed concepts:
1. Table Slider
    a. Jonathan did not like the slider prototype due to the mechanical aspects of the design
        i. Usage would require more force if unexpected friction when sliding is present.
        ii. Usage required a twisting motion to rotate and twist the camera, which proved to be uncomfortable.
2. Handlebars
    a. Proved to be unsafe and unstable
    b. Upon observation, we realized that if it failed, it would fall on his head
    c. Too complex, worst stability
3. Gimbal
    a. Has the best degrees of freedom

Verification Results

	Handlebars	Gimbal	Notes
Distance from eyes	30–50 As the handle pushed	Can be 35 cm exactly (by design)	Verification by analysis - calculated via simulation
Degrees of Freedom	～40 degrees forwards One axis (if the second axis is added, the system will be much unstable)	~270 degrees rotation (not 360 since view of monitor is not usable footage) ~120 degrees tilt	Verification by in-person meeting and testing
Force to operate		~ 2 N (weight of iPhone XR)	Verification by testing
Total Assembly Weight			Verification by analysis - calculated in CAD
Vibration/Stability	Visibly shakes for 0.3-0.5 seconds	Negligible (used recording of GoPro-similar camera)	Verification by testing for gimbal, by inspection for handlebars

List of Design Decisions Made

DIVERGING - Many components of possible design concepts (overhead, reinforcement, projection, supportive platform, VR-type glasses), but they had a lot of overlap and potential to mix-and match. A morph chart helped us scope these components into MC and RC



MVC GIMBAL

1. **Arm => Gimbal**
   Better stability and sustainability (lies on the table rather than hanging off it)
2. **Button layout**
   Four buttons in a diamond shape, arrows indicating rotation and tilt
   Chose this over a lined up button layout with arrows drawn on the buttons
   Large (visibility), intuitive/simple (learnability) and stable

HANDLEBAR

1. **Hanging Rod => Handle**
   Designed for stability and intuitive control (learnability)
2. **Control Panel => Two buttons**
   Design decision involved gearing towards simplicity

TABLE SLIDER

RESEARCH

Pan/Tilt

Why move the camera at all?

There are many reasons to move a camera. It can be used to:

1. Reveal information, or hide something.
2. Add energy to a scene, or uncertainty.
3. Get a better look at things, or follow an actor, discovering the scene along with him or her.
4. Move away from the actor, detaching yourself from him or her.

When used correctly camera movements are one of the most powerful elements a storyteller has in his or her arsenal.

Let's see what the different types of camera movements are, why they are used, and what tools you can use to make those moves.

1 Pan and tilt movement

The camera is fixed in a position but looks right or left, or up or down. This is the simplest type of camera movement – and the earliest.

You can show an actor walking from left to right, revealing the scene, or you can point from shoe to head, revealing their attire.

https://wolfcrow.com/the-5-different-types-of-camera-movement/